

Statistical Modeling

DATA.STAT.740

Lecture 10

Henri Pesonen 2025-2026

Nonparametric regression

Sometimes we need more complex mechanisms for describing the associations of predictors and outcome variables. For normal model with additive noise:

$$y_i = h(x_i) + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2)$$

Analogous formulation can be made also for other GLMs

In **nonparametric regression** we consider the structure of the function $h(x)$ unknown and approximate it with a linear combination of **basis functions**

$$h(x_i) \approx \phi^T(x_i)\beta, \quad \phi^T(x_i) = [\phi_1(x_i), \phi_2(x_i), \dots, \phi_m(x_i)]$$

Examples: polynomial regression, Fourier basis regression, spline regression
- Continuous predictors

Polynomial regression

When the association of a predictor x_j to the $g(E(y_i))$ is curved rather than straight, it is possible to use polynomial regression to model this

Polynomial regression (of a single predictor x_j) can be written as

$$g(E(y_i)) = \beta_0 + \beta_1 x_j + \beta_2 x_j^2 + \cdots + x_j^P$$

Notice that the effect of the predictor x_j is now nonlinear **but** expressed as a linear combination of the components

Polynomial regression

In polynomial regression, the predictors are strongly recommended to be normalized (before the transformation!) because the scale of the components blow up

$$x \in (0, 100) \rightarrow x^2 \in (0, 10000) \rightarrow x^3 \in (0, 1000000)$$

Polynomial regression is a simple approach for modeling curved responses, but care has to be taken

- too many components with respect to the dataset size can cause problems
 - the curve fits too well to the observations but doesn't generalize

- possibly erratic performance close to the smallest and largest predictor values

R-example!

Fourier basis regression

Fourier basis regression is suitable when there is a periodic component in the outcome. T is the period and we select K harmonics

$$\phi(x_i) = \begin{bmatrix} 1 \\ \sin\left(\frac{2\pi \cdot 1}{T} \cdot x_i\right) \\ \cos\left(\frac{2\pi \cdot 1}{T} \cdot x_i\right) \\ \sin\left(\frac{2\pi \cdot 2}{T} \cdot x_i\right) \\ \cos\left(\frac{2\pi \cdot 2}{T} \cdot x_i\right) \\ \vdots \\ \sin\left(\frac{2\pi \cdot K}{T} \cdot x_i\right) \\ \cos\left(\frac{2\pi \cdot K}{T} \cdot x_i\right) \end{bmatrix}$$

Fourier basis regression

Advantages

- Perfect for periodic data

- Produces extremely smooth fits

- Basis functions are orthogonal (for evenly spaced data) → stable estimation

- Avoids oscillatory edge artifacts of high-degree polynomials

- Great interpretability in frequency domain (e.g., dominant cycles)

Disadvantages

- Not ideal for non-periodic relationships

- Global basis → local anomalies affect the whole curve

- Harder to model sharp changes or discontinuities

- Requires choosing a truncation level K

R-example!

Spline regression

Spline regression is an extension of polynomial regression that fits piecewise polynomial functions to data.

Instead of forcing one global polynomial curve (like quadratic or cubic) across the whole range of x , in spline regression modelling we use different polynomial shapes in different regions predictor space

These regions are separated by points called **knots** ($\tilde{x}_j$)

$$\min(x) = \tilde{x}_0 < \tilde{x}_1 < \tilde{x}_2 < \dots < \tilde{x}_K = \max(x)$$

Spline regression

Common choice for spline regression are cubic splines

Essential there are third degree polynomial defined on each of the intervals defined by the knots

Third degree polynomial have a good balance of flexibility

Model is $h(x_i) = \phi^T(x_i)\beta$

where

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ [(x - \tilde{x}_1)_+]^3 \\ [(x - \tilde{x}_2)_+]^3 \\ \vdots \\ [(x - \tilde{x}_K)_+]^3 \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \theta_1 \\ \vdots \\ \theta_K \end{bmatrix} \quad \text{and}$$

$$(t)_+ = \max(t, 0)$$

This means that $(x - \tilde{x}_j)_+^3$
“turns on” only
when $x > \tilde{x}_j$

Spline regression

The basis $\phi(x)$ creates extra curvature after each knot and preserves

- the continuity of the function
- the continuity of the first derivative
- the continuity of the second derivative

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ [(x - \tilde{x}_1)_+]^3 \\ [(x - \tilde{x}_2)_+]^3 \\ \vdots \\ [(x - \tilde{x}_K)_+]^3 \end{bmatrix}$$

The model is constructed from the basis functions

$$\mathbf{y} = \Phi\beta + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma^2 I)$$

and the maximum likelihood estimator is

$$\Phi = \begin{bmatrix} \phi(x_1)^T \\ \phi(x_2)^T \\ \vdots \\ \phi(x_N)^T \end{bmatrix}$$

$$\hat{\beta} = (\Phi^T \Phi)^{-1} \Phi^T \mathbf{y}$$

Spline regression

$$h(x_i) \approx \phi^T(x_i)\beta, \quad \phi^T(x_i) = [\phi_1(x_i), \phi_2(x_i), \dots, \phi_m(x_i)]$$

Penalized estimation methods are often used to estimate β

It reduces the “wiggleness” of the curve

For cubic splines the estimate is often found as a solution to

$$\min_{\beta} (\mathbf{y} - \Phi\beta)^T (\mathbf{y} - \Phi\beta) + \lambda \int_{\tilde{x}_0}^{\tilde{x}_K} (h^{(2)}(x))^2 dx$$

where $h^{(2)}(x)$ is the second derivative of $h(x)$

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ [(x - \tilde{x}_1)_+]^3 \\ [(x - \tilde{x}_2)_+]^3 \\ \vdots \\ [(x - \tilde{x}_K)_+]^3 \end{bmatrix}$$

Spline regression

$$h(x_i) \approx \phi^T(x_i)\beta, \quad \phi^T(x_i) = [\phi_1(x_i), \phi_2(x_i), \dots, \phi_m(x_i)]$$

Use
$$h^{(2)}(x) = \frac{d^2\phi(x)}{dx^2} = \left(\phi^{(2)}(x)\right)^T \beta$$

and then

$$\begin{aligned} \lambda \int_{\tilde{x}_0}^{\tilde{x}_K} (h^{(2)}(x))^2 dx &= \lambda \int_{\tilde{x}_0}^{\tilde{x}_K} \left(\left(\phi^{(2)} \right)^T \beta \right)^2 dx = \lambda \int_{\tilde{x}_0}^{\tilde{x}_K} \left(\left(\phi^{(2)} \right)^T \beta \right) \left(\left(\phi^{(2)} \right)^T \beta \right) dx \\ &= \lambda \beta^T \int_{\tilde{x}_0}^{\tilde{x}_K} \phi^{(2)} \left(\phi^{(2)} \right)^T dx \beta \\ &= \lambda \beta^T R \beta \end{aligned}$$

$$\phi(x) = \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \\ [(x - \tilde{x}_1)_+]^3 \\ [(x - \tilde{x}_2)_+]^3 \\ \vdots \\ [(x - \tilde{x}_K)_+]^3 \end{bmatrix}$$

and minimization problem can be expressed as
$$\min_{\beta} (\mathbf{y} - \Phi\beta)^T (\mathbf{y} - \Phi\beta) + \lambda \beta^T R \beta$$

and it has a solution
$$\hat{\beta} = (\Phi^T \Phi + \lambda R)^{-1} \Phi^T \mathbf{y}$$

called **generalized ridge estimator**

R-example!

End of Lecture 10